



RESTORATIONESSENTIALS

Factory-Correct Restoration Reference

1970 Oldsmobile 442 Restoration Guide

Model Year 1970

Model	Oldsmobile 442 (Holiday coupe, Sports Coupe, conv.)
Platform	GM A-body
Engines	455 V8 (365 hp) / W-30 455 (370 hp)
Transmissions	Muncie 4-speed / Turbo Hydra-matic 400
Notable	Peak year; W-30 + Indy pace-car replicas

Every specification in this guide was checked against factory literature, marque references, period road tests, and documented survivor cars. Where sources disagree or the record is silent, this guide says so rather than guessing.

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Chapter 1 — Oldsmobile and the 4-4-2 Idea

Oldsmobile built the 4-4-2 the way it built its reputation: on hardware. When the badge appeared in 1964 on the Cutlass, the digits stood for the recipe — a four-barrel carburetor, a four-speed manual, and dual exhausts. By 1968 the 4-4-2 was a model in its own right, and by 1970 it stood at the top of the GM A-body performance ladder: the standard engine was the 455, the optional W-30 was the strongest street engine Oldsmobile ever sold, and the factory put 4-4-2s at the front of the Indianapolis 500 as pace cars. The W-Machines, as Oldsmobile advertised them, were marketed with a wink — 'Dr. Oldsmobile' and his W-30 — but the cars underneath the advertising were serious, carefully specified machines.

That is why this guide is organized the way it is. A 1970 442 is not valuable because it is old; it is valuable because of what it is — and 'what it is' lives in the VIN, the cowl tag, the casting numbers, and the broadcast card under the rear seat. Restore the identity first and the metal second, and the car will always be what its paperwork says it is.

Chapter 2 — What Makes 1970 Different

1970 is the peak. The 455 became the standard 442 engine this year at 365 horsepower and 500 lb-ft of torque — the strongest standard rating the model ever carried — and the W-30 package took the same engine to 370 horsepower with better-breathing heads, a performance camshaft, and Oldsmobile's 'Select Fit' hand-matched internal parts. Compression stood at 10.25:1 on the standard 455; period and auction references cite 10.5:1 for the W-30 — where sources differ this guide says so rather than splitting the difference.

Styling changed with the power: 1970 cars wear their own one-year look, and Oldsmobile built the 442 in three body styles — the Holiday hardtop coupe, the convertible, and the pillared Sports Coupe. This was also the Indianapolis year: 626 pace-car replicas were built, 268 of them on the 442 and 358 on the Cutlass Supreme (see Chapter 9). After 1970, falling compression and tightening emissions rules pulled the ratings down year by year, which is why restorers treat the 1970 car as the high-water mark.



Chapter 3 — VIN and Cowl-Tag Decoding

The VIN is on a plate at the base of the windshield on the driver's side. A 1970 442 reads like this worked example: **344870M100001**. Read it position by position:

- **3** — Oldsmobile division.
- **44** — the 442 series. This pair is the first identity check: a 1970 Cutlass carries a different series code, and no trim package turns a Cutlass VIN into a 442 VIN.
- **87 / 67 / 77** — body style: 87 Holiday hardtop coupe, 67 convertible, 77 Sports Coupe.
- **0** — model year 1970.
- **Plant letter** — M is Lansing, Michigan; E is Linden, New Jersey; Z is Fremont, California.
- **Sequential number** — the car's build sequence. Since 1968, federal rules required this same sequential number to be stamped into the engine block and the transmission case. On a numbers-matching 1970 442, all three stamps agree — that triangle is the foundation of the car's identity.

The cowl tag (on the firewall, driver's side) carries the build data. **ST 70** marks the 1970 style line, followed by the body-style number (34467 convertible, 34487 Holiday coupe). Trim codes you will meet: **930** black, **937** ivory, **934** gold, **973** blue. Paint codes include **75** Matador Red, **55** Galleon Gold, **14** Platinum, **20** Azure Blue, and **73** Rallye Red, a special-order color. The trim and paint codes on the tag — not the color the car wears today — are what the car was.

Chapter 4 — The 455: Standard and W-30

Every 1970 442 left the factory with a 455. In standard form it was rated 365 horsepower and 500 lb-ft of torque at a 10.25:1 compression ratio — a big-port, under-stressed engine that made its power with torque rather than rpm. The blocks are the 1968–72 455 'F' casting family shared with the big-car lines, which matters for interchange (Chapter 15). A four-barrel Rochester Quadrajet sits on top, and period press agreed the published ratings were conservative; this guide prints the factory ratings and leaves the dyno arguments to the dyno sheets.

The W-30 engine is the same 455, built tighter: 'F' casting high-compression cylinder heads (the standard 442 carries 'E' casting heads — the single fastest under-hood tell), an aluminum intake manifold with a specially calibrated Quadrajet, a high-performance camshaft, low-restriction exhaust, and Select Fit internals — pistons, bearings, and valve springs hand-matched at the factory. Rated output: 370 horsepower at 5,200 rpm and 500 lb-ft at 3,600 rpm.

Chapter 5 — The W-30 Package in Depth

Order code W-30 was a package, not an engine, and a correct car shows all of it:

- **W-25 Force-Air hood** — the fiberglass hood with twin scoops and a sealed air cleaner; ordered with the W-30 and also available on its own (see Chapter 13).
- **Red plastic inner fender liners** — the famous 'red fenders,' fitted to shed weight. Originals are brittle with age; reproductions exist and Chapter 16 covers the tells.
- **Aluminum intake and 'F' heads** (Chapter 4).
- **W-27 aluminum differential cover** on the Oldsmobile 12-bolt rear axle — ordered with the package on many cars.
- **Drivetrain calibration** — the 4-speed cars paired with 3.42:1 gearing and the Turbo Hydra-matic 400 cars with 3.23:1; period references note dealer-installed ratios up to 5.00:1, so a gear set that is not 3.42 or 3.23 is not automatically wrong — it is a question to document, not a defect to hide.



- **Anti-Spin rear axle standard** on the W-30, front disc brakes (unassisted on 4-speed cars), sway bars front and rear, and G70-14 raised-white-letter tires.
- **OW-code Turbo Hydra-matic 400** — the transmission code specific to W-30 automatics, and one of the strongest authenticity checks on an automatic car.

Production context: 3,100 W-30s were built for 1970 — 2,574 Holiday coupes and 264 convertibles, 96 of those convertibles with the 4-speed. Documentation is the hard part: no detailed build sheet is documented for the 1970 W-30, so the broadcast card (Chapter 3, under the rear seat) and the VIN-stamp triangle carry the proof. A W-30 without paper is a research project — price and restore it accordingly.

Chapter 6 — Transmissions: Muncie and Turbo Hydra-matic 400

Two gearboxes were offered. Manual cars use a Muncie 4-speed (close-ratio M21 units are documented in W-30 survivors; the transmission carries the VIN sequential stamp, so a swap announces itself). Automatic cars use the Turbo Hydra-matic 400 — the heavy-duty three-speed, with the OW code reserved for the W-30 and the Hurst Dual/Gate console shifter available under option W-26 on console cars. Restoration sequence for any of them: decode and record every stamp before disassembly, because the stamps — not the internals — are the identity.

Chapter 7 — Axles, Brakes, Suspension, and Wheels

The rear axle is the Oldsmobile-built 12-bolt (not the Chevrolet 12-bolt — covers and some internals differ, and Chapter 15 returns to this). Axle ratio codes are stamped on the axle; a documented 1970 W-30 4-speed, for example, carries axle code TO for its 3.91 Posi-Traction gearing with the W-27 aluminum cover. Option G80 is the Anti-Spin limited-slip; FE2 is the heavy-duty suspension with sway bars front and rear; JL2 is power front disc brakes. Wheels: the N66 Super Stock II styled steel wheel with G70-14 raised-white-letter tires is the signature 1970 combination. Brake work rule: identify the system from the option codes first — manual-disc, power-disc, and drum cars use different hardware, and ordering by year alone will eventually bite.

Chapter 8 — Paint, Trim, and Factory Stripes

The cowl tag is the authority (Chapter 3). Confirmed 1970 codes: 75 Matador Red, 55 Galleon Gold, 14 Platinum, 20 Azure Blue, and 73 Rallye Red as a special-order color — the pace-car replicas (Chapter 9) wear their own documented scheme. Trim: 930 black, 937 ivory, 934 gold, 973 blue. Hood striping and W-30 callouts were part of the factory presentation; before repainting, photograph and measure the surviving stripes on an original car wherever possible — stripe placement is judged against factory practice, not against another restored car.

Chapter 9 — Production Numbers and Rarity

Body style	Built	Notes
Holiday hardtop coupe	14,709	The volume 442
Convertible	2,933	Scarcest regular body style
Sports Coupe	1,688	Pillared coupe
Total 442	≈19,330	All body styles
W-30 package	3,100	2,574 coupes / 264 conv.
Indy pace-car replicas	626	268 on 442 / 358 on Cutlass Supreme

Read rarity the way the factory built it: a W-30 convertible with the 4-speed (96 built) sits at the top of the 1970 desirability ladder, and the pace-car replicas carry their own documented identity. Rarity claims beyond these



published splits — color-and-option combinations, for example — cannot be verified from surviving factory records; this guide does not print them, and you should not pay for them.

Chapter 10 — Rust-Prone Areas and Inspection Guide

The 442 is a GM A-body, and it rusts like one: inspect the cowl and windshield frame, the lower front fenders behind the wheels, the rear quarter panels and wheel openings, the trunk floor (water enters past the rear window and deck seals), the floor pans under the carpet, and the frame at the rear rails and body mounts. Convertibles add the top well and header bow. On any W-30, remember what is at stake: the fiberglass W-25 hood and red liners survive or they don't, and a rust repair that costs a numbers-matching car its original panels is a value decision, not just a metal decision. Probe with a magnet and a light, photograph everything, and write down what you find before any negotiation — Chapter 14 turns this into a checklist.

Chapter 11 — Restoration Sequence for the 1970 442

- **1. Document.** Photograph the car as found; record the VIN, cowl tag, engine/transmission stamps, and casting dates; look for the broadcast card under the rear seat before the interior comes out.
- **2. Decide the car's honest identity.** Matching-numbers 442, documented W-30, or a car with questions — the answer sets the budget and the standard.
- **3. Disassemble by system,** bagging and tagging hardware; the small 1970-only parts (trim, lamps, stripe details) go into labeled storage first because they are the hardest to replace.
- **4. Metal and frame first,** then drivetrain, then paint to the cowl-tag code, then trim to the tag code, then details (stripes, callouts, date-coded wear items).
- **5. Verify against the tag, not against taste** at every reassembly step.

Chapter 12 — Engine Specifications at a Glance

Item	455 (standard)	455 W-30
Factory rating	365 hp	370 hp @ 5,200 rpm
Torque	500 lb-ft	500 lb-ft @ 3,600 rpm
Compression	10.25:1	10.5:1 cited (see Ch. 4 note)
Cylinder heads	'E' casting	'F' casting
Induction	Rochester Quadrajets	Calibrated Quadrajets, aluminum intake
Internals	Production	Select Fit matched

Chapter 13 — Options and Factory Details

Code	What it is
W-25	Force-Air fiberglass hood (twin scoops)
W-26	Console with Hurst Dual/Gate shifter (auto)
W-27	Aluminum differential cover
W-30	The performance package (Chapter 5)
FE2	Heavy-duty suspension
G80	Anti-Spin rear axle (posi)
JL2	Power front disc brakes
U21	Rally Pac gauges with Tic-Toc-Tach
N66	Super Stock II styled wheels

Options prove out on the cowl tag and the broadcast card, not on the car's current equipment — fifty years is plenty of time for a Dual/Gate console to move between cars.

Chapter 14 — Buyer Inspection Checklist

- VIN plate reads 344 + body code + 0 + plant letter; series digits are 44, not a Cutlass code.
- Engine and transmission stamps match the VIN sequential (1968-on federal stamping rule).
- Cowl tag paint/trim codes match the car's documented colors.
- W-30 claims: 'F' heads, aluminum intake, W-25 hood, red liners, OW code on automatic cars, Anti-Spin — all present, or the price reflects what is missing.
- Broadcast card found (check under the rear seat) or its absence priced in; no 1970 W-30 'build sheet' should be promised by a seller.
- Rust map complete (Chapter 10) with photos.
- Axle code stamped and ratio verified by rotation count.
- Date codes on block/heads/intake precede the car's build date and sit close to it.

Chapter 15 — Casting Numbers and Date Codes

Casting numbers identify the part family; date codes place it in time; the VIN stamp ties it to the car. On one thoroughly documented 1970 W-30 4-speed, the recorded stampings are: block casting 396021 F ('F' = the 1968–72 455 family) with the VIN sequential stamped on the block, head casting 404438, intake 406115, Quadrajet 7040256, distributor 1111977, Muncie casting 3925660 with the VIN stamp, and axle code TO (3.91 Posi) under the W-27 cover. Treat that set as the format to look for — the exact digits on your car are read off your car. A correct date sits before the car's build week and near it; a part dated after the build, or months before it, is a question to answer before money moves. Interchange honesty: the 455 'F' block family is shared across 1968–72 big Oldsmobiles, but heads, intakes, and calibration parts are application-specific — the casting number, not the seller's word, decides.

Chapter 16 — Reproduction vs. Original: Tells That Matter

The W-30's signature parts are all reproduced, which is good news for drivers and a trap for authenticity claims. The red inner liners: originals are aged, chalked, and brittle at the mounting holes; crisp new plastic is a replacement until proven otherwise. The W-25 hood: factory fiberglass carries period mold character; the scoop openings and underside finish differ from reproduction runs. The W-27 cover and the 'F' heads cannot be faked cheaply — which is why Chapters 5 and 15 put the verification weight there and on the VIN stamps. None of this makes a reproduction part bad; it makes an undisclosed one a misrepresentation. Document what you install.

Chapter 17 — Critical Fasteners: Sequence and Sources

Torque values are factory-manual data, and this guide does not print a number it cannot trace to the manual. What it prints is the discipline that makes the manual's numbers work: chase and clean every thread, lubricate fasteners the way the manual specifies for the value you are using, work in the printed pattern in at least three steps, and re-check after the first full heat cycle where the manual calls for it. Take the values themselves from the 1970 Oldsmobile Chassis Service Manual's specification section — the same manual listed in Sources — and write the values you used into your build sheet. A number remembered from a forum is not a torque spec; a number copied from the factory page is.

Recommended for this restoration

Shop exact-fit supplies on Amazon

- 1970 Oldsmobile 442 factory service manual
- Oldsmobile 455 engine rebuild kit
- Oldsmobile 442 W-30 reproduction parts
- 1970 Oldsmobile Cutlass 442 parts catalog

As an Amazon Associate, RestorationEssentials earns from qualifying purchases made through these links.

Sources Consulted

Every specification in this guide was checked against published references before printing. Where sources disagree, this guide says so in the text rather than picking a number silently. The references below are the backbone of the technical chapters:

- **Oldsmobile factory literature.** the 1970 Oldsmobile Chassis Service Manual and factory sales literature — the authority for torque values, specifications, and option codes
- **Published marque references.** period road tests and marque histories of the W-Machines, including Publications International's muscle-car references (HowStuffWorks) for the W-30 package content and ratings
- **Auction research files.** RM Sotheby's and Mecum lot research for W-30 production splits and the Indianapolis pace-car replica counts
- **Marque data compilations.** Oldsmobile 442 production and decode data as compiled by speedofdaily.com and GM Authority from factory-derived records



- **Documented survivor cars.** numbers-recorded 1970 W-30 cars in dealer and auction listings (classiccars.com and published video documentation), used for casting-number and stamp formats and labeled as examples in the text

A note on numbers. Factory literature, period road tests, and marque historians do not always agree — production totals, horsepower ratings, and paint codes all have documented conflicts. This guide uses the best-supported figure in each case and flags the conflict where one exists, so you can judge for yourself.



Thank You

Thank you for choosing this RestorationEssentials guide. Whether this is your first 1970 Oldsmobile 442 or your fiftieth, you picked one of the great ones, and the fact that you are doing your homework before you turn a wrench puts you ahead of most of the field.

These cars survived because people like you refused to let them disappear. Every correct fastener, every properly decoded tag, every hour spent getting the details right keeps the history honest for the next owner and the next generation of enthusiasts.

The 1970 442 is the year everything lined up — the big 455, the W-30 at full song, and a pace car at Indy. Getting one right starts with the numbers, and now you have them.

If this guide helped your build, we would love to hear about it. Good luck with the restoration, and enjoy every mile.

— The RestorationEssentials Team